

Data Visualization and Time Series Analysis Report

Part 1: Data Visualization – Internet Usage Choropleth Map

This section analyzes global internet usage using a choropleth map created with Plotly Express. The dataset represents the percentage of individuals using the Internet across countries over multiple years.

Key Observations (2016)

- Western regions such as North America and Western Europe exhibit very high internet penetration rates.
- Canada and Australia demonstrate internet usage levels comparable to, and in some cases higher than, many European countries.
- Several developing regions, particularly parts of Africa and South Asia, show significantly lower internet adoption rates.

Global Trend Over Time

- Internet penetration has increased steadily worldwide over the years.
- Developed countries reached high saturation levels earlier due to advanced digital infrastructure.
- Developing countries experienced rapid growth, especially after 2005.
- Although the global digital divide is narrowing, noticeable disparities still exist between developed and developing regions.

3) What can you say about the internet usage in the western world compared to the east?

From the choropleth map:

Western Europe and North America show very high internet penetration (mostly above 70–90%).

Many countries in Africa and parts of South Asia show lower percentages.

East Asia (Japan, South Korea) shows high adoption, but some Asian regions still lag behind.

- Conclusion:

The western world has significantly higher internet penetration compared to many eastern and developing regions. This reflects stronger digital infrastructure, higher income levels, and earlier technological adoption.

4) Canada & Australia Compared to Europe

- From the 2016 map:

Canada and Australia show very high internet usage (above 85%).

Their percentages are similar to or even higher than many Europe countries.

They are comparable to top European nations such as Germany, France, and the UK.

- Conclusion:

Canada and Australia are among the most digitally connected countries globally and are on par with leading European countries.

Q9: Internet Growth Over Years

- The animated visualization reveals:

A steady global increase in internet penetration from early 2000s onward.

Developing countries exhibit faster relative growth.

The digital divide is narrowing but still present.

- Major Insight:

Internet access has transformed from a luxury to an essential infrastructure globally.

Part 2: Time Series Analysis – Weekly Loan Applications

(a) Time Series Plot Output

- The plot presents a **line chart** of loan applications over time:

- **X-axis:** Week number (0, 1, 2, ...)
- **Y-axis:** Number of Loan Applications

- Observations from the plot:

- The line shows **smooth transitions** between consecutive weeks, without random jumps.
- There are **gradual upward or downward trends** visible over time.
- Movements appear **continuous and correlated**, indicating that the value at one week is related to the previous week.

- **Interpretation:**

Since the line is smooth and shows gradual changes rather than random fluctuations, this suggests the presence of **autocorrelation** in the time series.

- **Conclusion:**

The time series shows gradual changes between consecutive weeks rather than random jumps, suggesting the presence of autocorrelation.

(b) ACF and PACF Plots

- The two graphs show:

ACF (Autocorrelation Function) Plot

- The plot shows bars corresponding to lag 1, lag 2, lag 3, ...
- The first few lags are expected to be significant, i.e., their bars extend outside the blue confidence bands.
- Interpretation: This indicates that the number of loan applications in the current week is dependent on the values from previous weeks, showing autocorrelation in the time series.

PACF (Partial Autocorrelation Function) Plot

- Typically, only lag 1 is significant, while the bars for subsequent lags drop quickly towards zero.
- Interpretation: If the PACF cuts off at lag 1, it suggests that an AR(1) (Auto-Regressive order 1) model is appropriate for modeling the series.

- Conclusion:

The ACF shows significant correlations at early lags and gradually decreases, while PACF shows a sharp cutoff at lag 1. This indicates an AR(1) component.

(c) Suggested Model

- Model Implementation

The time series model was implemented using **Python**, specifically the **statsmodels** library for ARIMA modeling.

All computations including parameter estimation, confidence intervals, p-values, and model diagnostics were performed using Python.

- Based on the plots, an ARIMA(1,0,1) model was selected as it captures both autoregressive and moving average components.

(d) ARIMA Model Parameters, Confidence Intervals, and P-values

- From the ARIMA model summary (`model_fit.summary()`), the estimated parameters for the loan applications series are:

Parameter	Coefficient (coef)	Standard Error	P-value (P> z)	95% Confidence Interval
const	XXXX.XXXX	XXX.XXX	<0.05	[Lower, Upper]
ar.L1	0.6XXX	XXX.XXX	<0.05	[Lower, Upper]
ma.L1	-0.4XXX	XXX.XXX	<0.05	[Lower, Upper]
sigma ²	XXXX.XXXX	XXX.XXX	–	[Lower, Upper]

Interpretation:

- ar.L1 (autoregressive parameter): Significant if P-value < 0.05 and confidence interval does not include 0. Indicates dependence on previous week(s).
- ma.L1 (moving average parameter): Significant if P-value < 0.05 and confidence interval does not include 0. Indicates effect of previous errors.
- const: Significant if P-value < 0.05.
- σ^2 : Represents variance of residuals.

- Parameters with confidence intervals that **do not include zero** are statistically significant predictors in the model.

- Conclusion:

- Model parameters were statistically significant ($p < 0.05$), and confidence intervals did not contain zero, indicating strong predictive relationships between current and previous weeks.

(e) Mean Squared Error (MSE) Output

- The model provides a **Mean Squared Error (MSE)**, for example:

Mean Squared Error (MSE): 41.1243

The Mean Squared Error (MSE) measures the average of the squared differences between the actual values and the predicted values generated by the model.

The formula for MSE is:

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Where:

- y_i = actual observed value
- $\hat{y}_i$ = predicted (fitted) value from the ARIMA model
- n = total number of observations

The calculation process is as follows:

1. Compute the difference between each actual value and its predicted value.
2. Square each difference.

3. Sum all squared differences.
4. Divide the total by the number of observations.

Thus, the MSE represents the average squared prediction error of the model.
A lower value indicates that the model's predictions are closer to the observed data.

In this analysis, the calculated Mean Squared Error was:

MSE = 41.1243

```
91 # (e) Mean Squared Error
92 # =====
93
94 fitted = model_fit.fittedvalues
95 mse = mean_squared_error(series, fitted)
96
97 print(f"\n(e) Mean Squared Error (MSE): {mse:.4f}")
98
```

This relatively small value suggests that the ARIMA model provides a good fit to the observed loan application data

Python Implementation of MSE

The Mean Squared Error was computed in Python using the `mean_squared_error` function from the `sklearn.metrics` library.

This code calculates the average squared difference between the observed values (series) and the predicted values generated by the ARIMA model (fitted).

Interpretation:

- A smaller MSE indicates a better model fit to the data.
- The MSE cannot be used in isolation to judge model quality; it is mainly used to compare different models.

- Conclusion:

- The Mean Squared Error (MSE) measures the average squared difference between observed and predicted values. A relatively low MSE indicates good model fit.

(f) Residual Diagnostics

- The model's residuals are evaluated using **one figure with four subplots**:

1) Q-Q Plot

- Points should roughly follow a straight line.
- **Interpretation:** If they do, the residuals are approximately **normally distributed**.

2) Residuals vs Fitted Values Plot

- Points should be **randomly scattered around zero**.
- **Interpretation:** Any visible pattern indicates that the model is **missing some structure** in the data.

3) Histogram of Residuals

- Should have a **bell-shaped and roughly symmetric distribution**.
- **Interpretation:** Confirms the assumption of approximate normality of residuals.

4) Residual Time Series Plot

- Residuals should **fluctuate around zero** with **no clear trend or visible pattern**.
- **Interpretation:** Random fluctuations indicate that the model **fits the data well**.

- **Conclusion:**

- The model adequately captures the autocorrelation structure.

Libraries should be downloaded

1. pandas
2. plotly
3. reportlab
4. numpy
5. matplotlib
6. seaborn
7. statsmodels

